

FORT WAYNE INTL, IN, USA

WMO: 725330

Lat: 40.971N

Lon: 85.206W

Elev: 791

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	4.5	-7.9	3.7	-0.4	-2.4	5.0	5.4	29.6	25.9	26.9	26.2	9.8	260	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	90.7	74.4	88.1	73.0	85.5	71.7	77.6	87.0	75.8	84.1	74.3	82.0	10.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	133.8	83.1	73.2	126.9	81.1	71.7	120.4	79.0	41.5	86.8	39.8	84.4	38.2	81.9	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	22.5	19.6	-7.1	94.3	7.1	3.5	-12.2	96.9	-16.4	98.9	-20.4	100.9	-25.6	103.4
			-7.6	80.1	6.9	2.0	-12.5	81.5	-16.6	82.7	-20.4	83.8	-25.5	85.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	25.2	28.8	38.6	50.4	61.3	70.8	73.6	71.6	65.1	53.5	41.3	30.8
	DBStd	19.21	12.27	11.33	10.78	9.14	8.47	6.61	5.44	5.04	7.11	8.77	9.38	10.48
	HDD50	2822	772	599	382	103	9	0	0	0	1	63	292	601
	HDD65	5948	1234	1015	819	444	175	23	3	7	85	370	712	1061
	CDD50	3202	3	4	29	116	360	624	733	670	456	172	30	5
	CDD65	853	0	0	1	7	61	197	271	213	89	14	0	0
	CDH74	7486	0	0	14	86	608	1711	2449	1670	809	138	1	0
	CDH80	2420	0	0	2	11	178	608	880	474	240	27	0	0
	WSAvg	9.1	11.2	10.7	10.5	11.0	9.4	8.0	6.8	6.1	6.7	8.6	9.8	10.2
	PrecAvg	37.0	2.2	2.0	2.8	3.5	4.0	4.2	3.7	3.5	2.8	2.7	2.9	2.7
Precipitation	PrecMax	54.5	5.4	6.9	5.7	7.1	10.2	12.0	11.0	9.7	6.8	8.9	8.0	7.6
	PrecMin	24.4	0.4	0.3	0.5	1.0	1.0	0.7	0.4	0.4	0.4	0.2	0.6	0.4
	PrecStd	6.4	1.3	1.2	1.1	1.5	2.0	2.1	2.0	1.8	1.5	1.6	1.5	1.3
	PrecStd	6.4	1.3	1.2	1.1	1.5	2.0	2.1	2.0	1.8	1.5	1.6	1.5	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	62.8	75.3	81.6	89.3	93.6	94.5	91.0	90.6	83.7	70.6	61.4
		MCWB	55.6	55.0	62.3	64.5	68.9	74.3	76.5	76.0	72.2	68.3	58.5	57.9
	2%	DB	53.1	55.7	67.2	76.1	84.8	90.0	91.0	87.8	86.5	77.9	64.4	55.5
		MCWB	50.4	51.2	56.7	61.3	68.1	72.9	75.9	74.8	70.7	64.1	56.9	52.5
	5%	DB	47.6	49.3	61.0	71.7	81.0	86.6	87.9	85.1	82.3	72.9	60.4	50.1
		MCWB	44.9	45.1	53.6	58.8	66.7	71.6	73.9	73.0	68.6	62.5	54.7	47.1
	10%	DB	40.9	43.2	55.8	66.8	76.6	83.3	85.0	82.4	78.7	68.5	56.1	44.6
		MCWB	38.5	39.9	50.0	56.1	64.7	70.1	72.4	71.5	66.7	60.0	51.0	41.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.7	57.8	63.7	68.1	73.5	77.8	80.3	79.2	76.3	71.3	62.2	58.2
		MCDB	58.0	60.4	74.6	78.1	82.8	88.9	89.9	87.9	85.8	80.1	67.5	61.3
	2%	WB	51.2	51.6	58.8	63.9	71.0	75.6	78.1	77.1	73.5	67.5	58.9	52.7
		MCDB	52.7	54.3	65.3	72.6	80.9	85.5	87.5	85.2	81.7	73.9	62.6	54.8
	5%	WB	45.5	45.4	54.9	60.6	68.8	73.8	76.3	75.1	71.3	64.4	55.3	46.8
		MCDB	47.8	49.2	60.3	68.4	77.4	83.0	84.8	81.7	78.1	70.7	59.9	49.4
	10%	WB	38.6	40.3	50.1	57.6	66.6	72.1	74.5	73.3	69.1	61.1	51.2	42.0
		MCDB	40.7	43.3	55.7	65.9	74.4	80.3	82.2	79.7	75.5	67.4	55.5	44.4
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.7	17.7	20.9	20.5	19.4	19.7	19.4	21.5	19.7	15.9	12.8
		MCDBR	17.3	20.8	25.5	27.4	25.6	23.3	21.9	21.5	25.7	25.4	21.8	18.1
		MCWBR	15.4	16.6	17.4	15.9	13.0	10.1	9.5	10.4	12.1	14.2	15.7	15.7
	5% WB	MCDBR	17.1	19.9	23.7	23.8	21.5	19.7	19.2	18.4	20.5	21.2	19.2	16.9
		MCWBR	15.8	17.0	17.3	16.0	12.3	10.0	9.5	10.0	11.0	13.6	15.5	15.3
Clear-Sky Solar Irradiance	taub		0.304	0.333	0.350	0.408	0.453	0.445	0.446	0.447	0.410	0.356	0.323	0.302
	taud		2.390	2.296	2.345	2.191	2.109	2.184	2.198	2.179	2.255	2.401	2.448	2.433
	Ebn at Noon		269	276	285	274	263	265	263	259	261	265	261	261
	Edh at Noon		26	33	36	44	49	46	45	45	39	30	25	23
All-Sky Solar Radiation	RadAvg		502	785	1122	1477	1708	1917	1929	1707	1394	937	603	421
	RadStd		50	96	96	134	155	133	105	88	114	87	80	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page